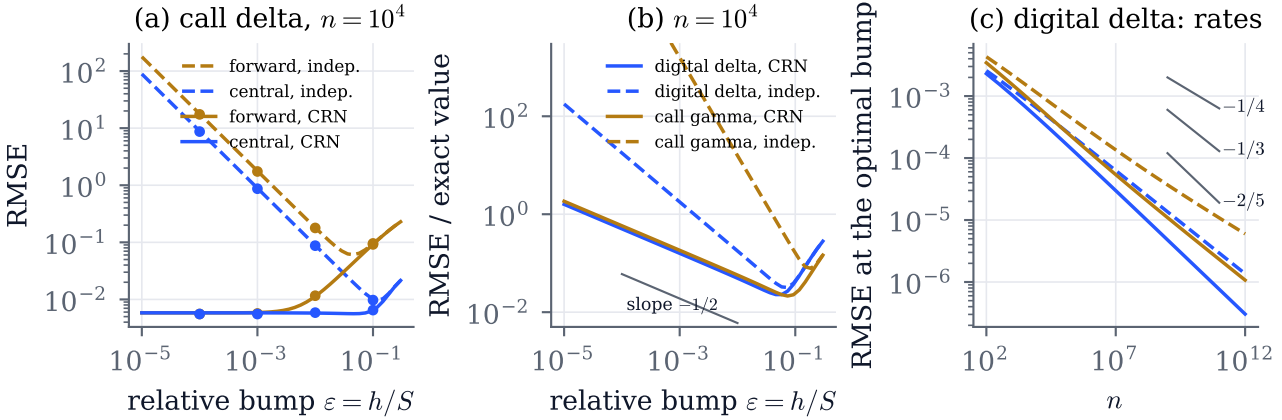
(a) call delta, $n = 10^4$
RMSE
forward, indep.
central, indep.
forward, CRN
central, CRN
10^{2}
10^{1}
10^{0}
10^{-1}
10^{-2}
10^{-5}
10^{-3}
10^{-1}
relative bump $\varepsilon = h/S$
(b) $n = 10^4$
RMSE / exact value
digital delta, CRN
digital delta, indep.
call gamma, CRN
call gamma, indep.
slope −1/2
10^{2}
10^{0}
10^{-2}
10^{-5}
10^{-3}
10^{-1}
relative bump $\varepsilon = h/S$
(c) digital delta: rates
RMSE at the optimal bump
−1/4
−1/3
−2/5
10^{-3}
10^{-4}
10^{-5}
10^{-6}
10^{2}
10^{7}
10^{12}
n